

Driving Repurchase Intent: Olist Regional Logistics & Loyalty Analysis

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Database Management: PostgreSQL

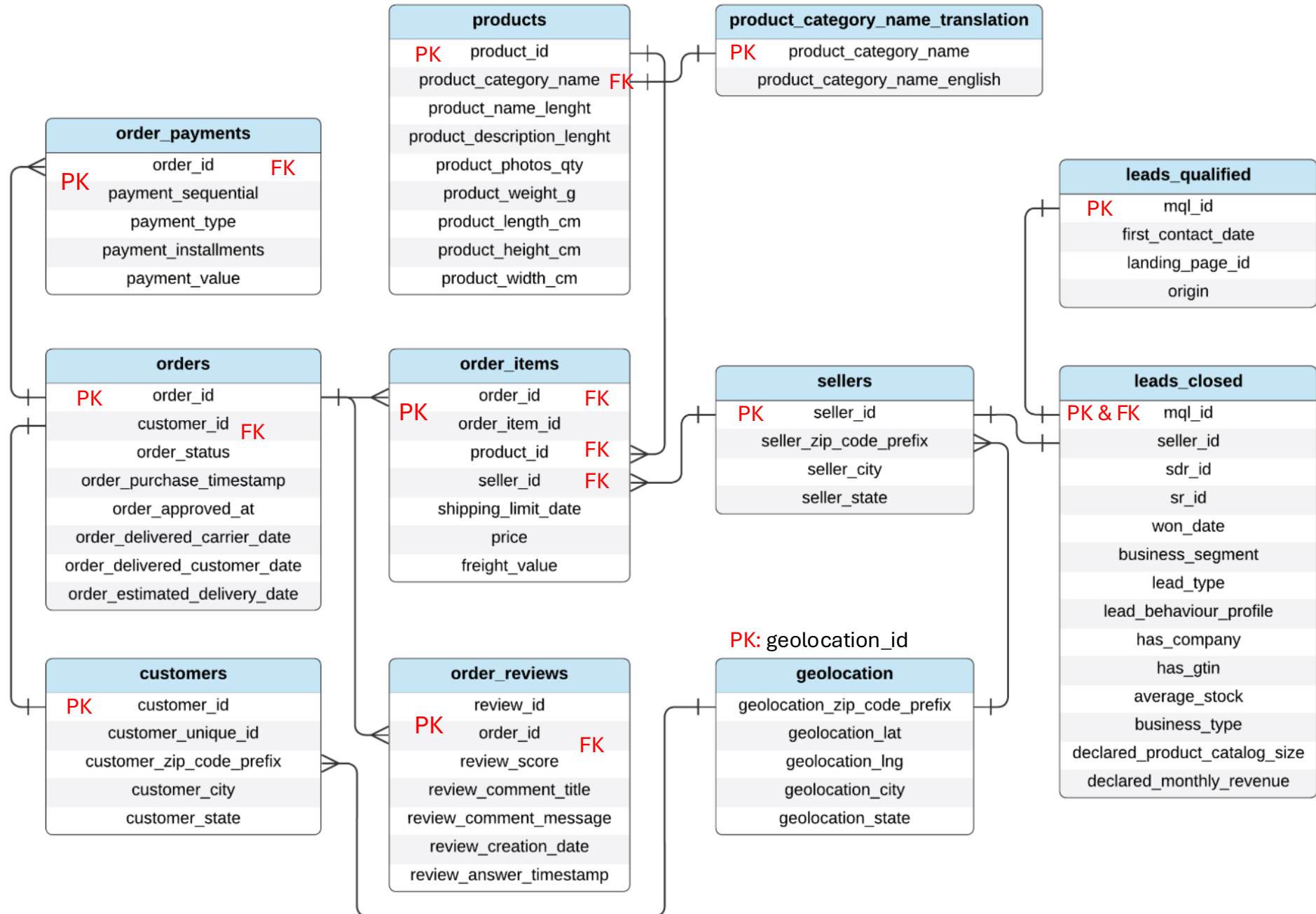
Data Visualization: Power BI

01. Dataset Overview & Research Objectives

Dataset Overview

- 100k Real Orders: Comprehensive commercial data spanning from 2016 to 2018 across multiple marketplaces in Brazil.
- Multi-Dimensional Features: Covers the entire e-commerce lifecycle including order status, pricing, payment methods, freight performance, and product attributes.
- Anonymized: Real commercial data anonymized for privacy.

DATABASE SCHEMA



Motivation:

Low Repeat Purchase and State-Level Variation

- Key Observation: Strong One-Time Purchase Pattern

3.12% Repeat Purchase Rate: Only 3k out of 96k unique customers reordered (Avg. 1.03 orders/customer).

total_unique_customers	one_time_customers	repeat_customers	one_time_customer_ratio_percentage	repeat_customer_ratio_percentage	total_orders	avg_orders_per_customer
93358	90557	2801	97.00	3.00	96478	1.03

Regional Retention Variance: Repeat purchase rates vary significantly by customer's main state.

retention_tier	number_of_states	included_states	total_customers	one_time_customers	repeat_customers	weighted_retention_rate_percentage	avg_state_retention_rate_percentage
>= 4%	2	AC, RO	307	293	14	4.56	4.80
3%-4%	5	GO, MT, RJ, RS, SP	58979	57119	1860	3.15	3.18
2%-3%	17	AL, AM, BA, DF, ES, MA, MG, MS, PA, PE, PI, PR, RN, RR, SC, SE, TO	32246	31351	895	2.78	2.53
< 2%	3	AP, CE, PB	1826	1794	32	1.75	1.73

State-Level Logistics & Customer Retention Analysis

Aim:

- **Analyze Retention Drivers:**

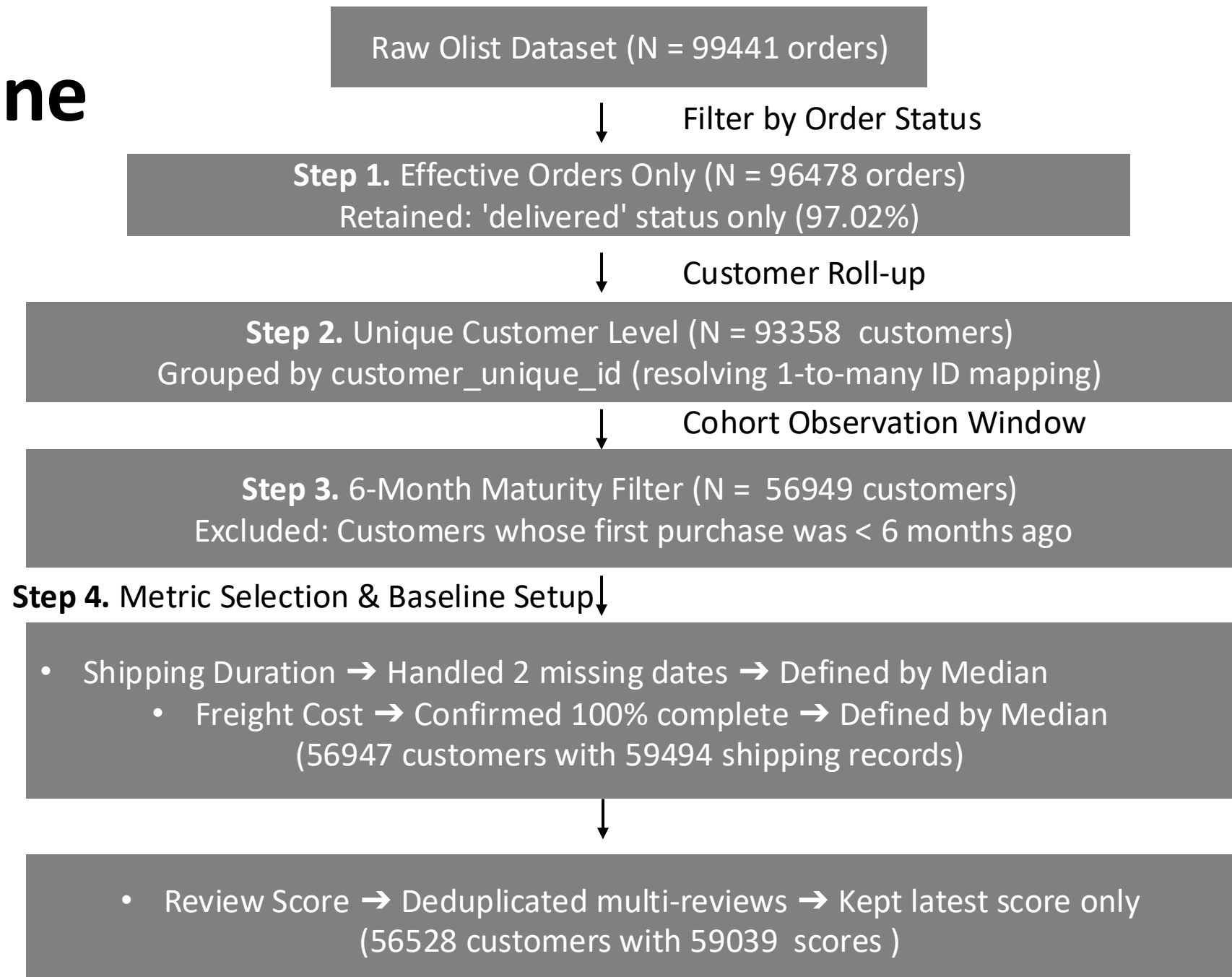
Evaluate if freight costs, cross-state shipping, delivery times, and review scores impact regional repeat purchases.

- **Optimize Operations:**

Identify logistical and satisfaction pain points to improve customer retention and strategy.

02. Data Preparation & Cohort Selection

Data Pipeline



Step 1. Defining "Effective Orders"

order_status	order_count	percentage
delivered	96478	97.02
shipped	1107	1.11
canceled	625	0.63
unavailable	609	0.61
invoiced	314	0.32
processing	301	0.30
created	5	0.01
approved	2	0.00

earliest_delivered_purchase_time	latest_delivered_purchase_time	delivered_orders
2016-09-15 12:16:38	2018-08-29 15:00:37	96478

Scope Limitation:

Analysis is strictly limited to successful, fulfilled orders (delivered status: 97.02%).

Purpose:

Eliminates statistical bias to ensure accurate measurement of true repurchase cycles and physical logistics duration.

Step 2. Customer Definition & Cohort Selection

Evaluated retention using customer_unique_id instead of customer_id to account for one-to-many mapping (multiple orders/IDs per unique user).

unique_customers	total_orders	total_items	items_per_order
93358	96478	110197	1.14

6-Month Cohort Filter:

Excluded users whose first purchase occurred less than 6 months ago.

Ensures newly acquired users have a sufficient observation window to exhibit natural repurchase behavior, preventing underestimation of true retention.

顧客首次購物時間點

first_purchase_quarter	new_customers_count	pct_of_total_new_customers
2016-Q3	1	0.00
2016-Q4	263	0.28
2017-Q1	4848	5.19
2017-Q2	8744	9.37
2017-Q3	11813	12.65
2017-Q4	16726	17.92
2018-Q1	19904	21.32
2018-Q2	18966	20.32
2018-Q3	12093	12.95

repurchase_interval_tier	repurchase_count	avg_days_within_tier	pct_of_total_repurchases
01. Within 1 Week	1106	0.7	35.45
02. 1-2 Weeks	185	10.3	5.93
03. 2 Weeks - 1 Month	273	21.4	8.75
04. 1-2 Months	359	44.4	11.51
05. 2-3 Months	225	73.6	7.21
06. 3-6 Months	457	131.4	14.65
07. Over 6 Months	515	283.4	16.51

Step 3. Cross-State Order Definition

order_composition_type	order_count	pct_of_total_delivered_orders
1. Single Seller Order	95203	98.68
2. Cross 2 Sellers	1216	1.26
3. Cross 3 Sellers	54	0.06
4. Cross 4 or More Sellers	5	0.01



geographic_distribution_type	order_count	pct_of_multi_seller_orders
1. Same State (Buyer & All Sellers in Same State)	341	26.75
2. Mixed States (Sellers Across States, Part Match with Buyer)	247	19.37
3. Cross States (All Sellers in Same State, But Different from Buyer)	425	33.33
4. Full Cross States (Buyer & All Sellers in Different States)	262	20.55

Multi-item orders frequently involve multiple sellers and distinct regional origins.

An order is classified as **"Cross-State"** if at least one item is shipped from a seller located in a different state than the buyer.

Step 4. Delivery Time / Shipping Duration

評估狀態/剔除原因	訂單筆數
1. Missing Carrier/Customer Date	3
2. Negative Shipping Duration	23

Purged 24 records with missing or unnormal delivered_date values to ensure timeline completeness.

total_valid_shipping_orders	min_days	q1_days	median_days	q3_days	p95_days	p99_days	max_days	negative_shipping_orders	over_90_days_orders
59496	-16.10	4.97	7.94	13.04	26.04	42.11	205.19	23	64

Identified severe right-skewness in shipping performance, where P95 is 29.27 days but the maximum extends to 209.63 days.

Adopted the median as the primary metric, ensuring robustness against extreme outliers and tracking true operational timelines.

Validation confirms extreme delays are scattered randomly across states with negligible ratios (mostly <0.5%), ruling out regional systemic bias.

Step 4. Freight Cost / Shipping Fee Filters

Confirmed 100% data completeness with zero null or negative records in the freight cost attribute

Detected a heavy tail in shipping fees; while 95% of orders cost under 54.78, extreme outliers spike up to 1794.96.

Standardized on the median as the central tendency measure, ensuring robustness against anomalous logistics pricing.

total_valid_freight_orders	min_freight	q1_freight	median_freight	q3_freight	p95_freight	p99_freight	max_freight	negative_freight_orders	over_150_freight_orders
59496	0.00	14.10	16.60	23.66	50.77	96.63	1002.29	0	172

Step 4. Review Score / Customer Rating Filters

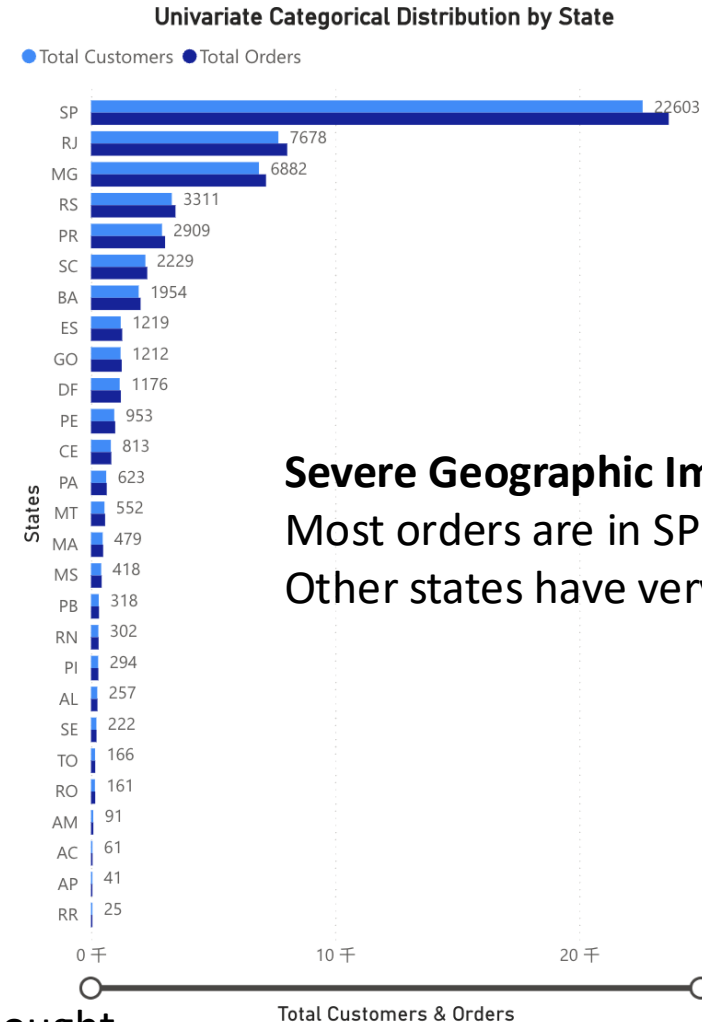
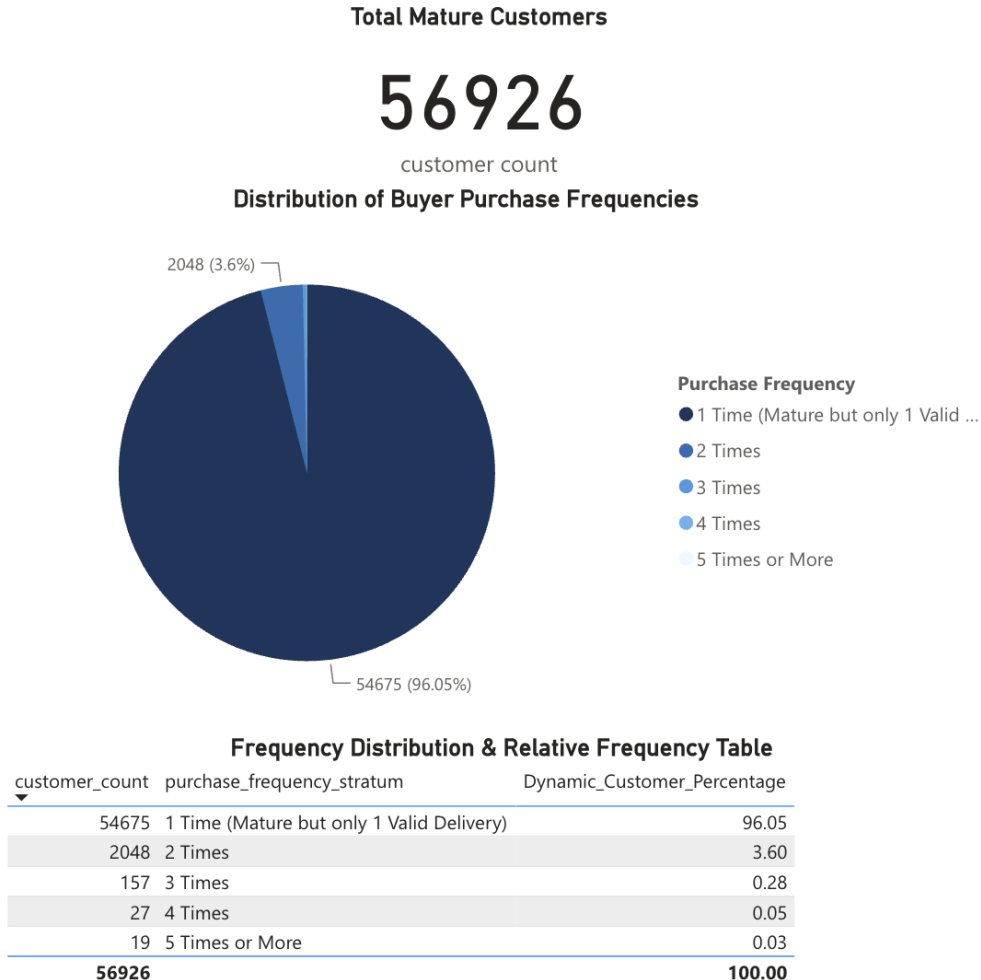
For orders with multiple reviews, only the latest score is retained to ensure a strict 1:1 order-to-review mapping.

review_score_tier	order_count	pct_of_total_orders
5 Stars	34236	57.54
4 Stars	11761	19.77
3 Stars	5151	8.66
2 Stars	1889	3.18
1 Star	6004	10.09
0 (No Review)	455	0.76

} Low Review Score
→ Exclude

03. Preliminary Analysis & Regional Retention Drivers

Preliminary Analysis: Geographic Disparity in Customer Density & Retention Rates

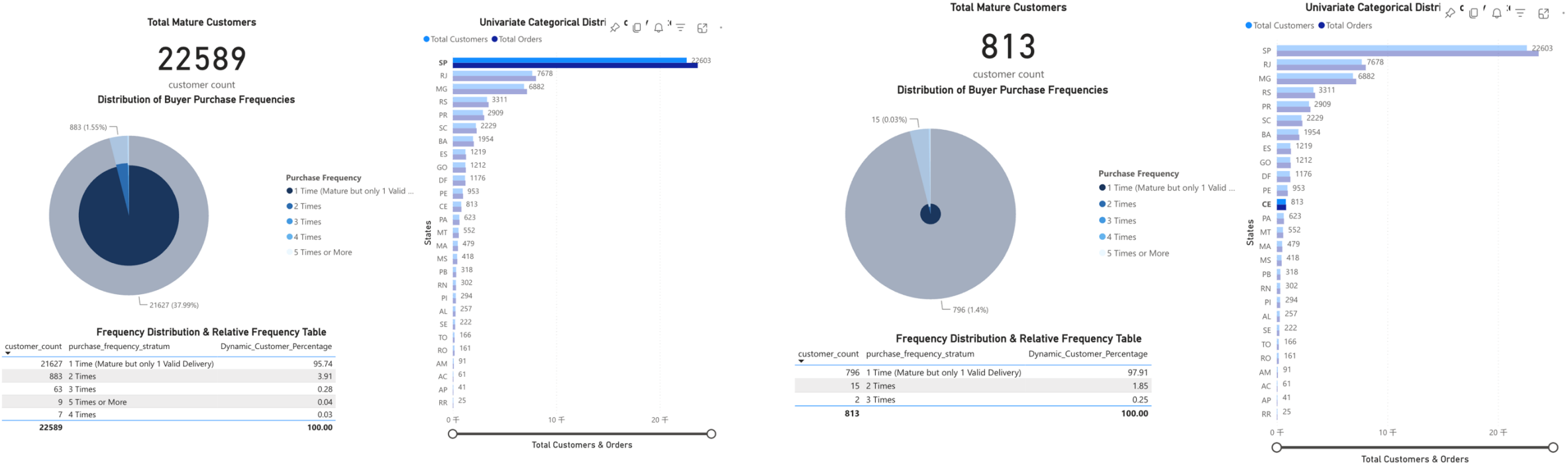


Severe Geographic Imbalance:
Most orders are in SP (22,603 orders).
Other states have very few orders.

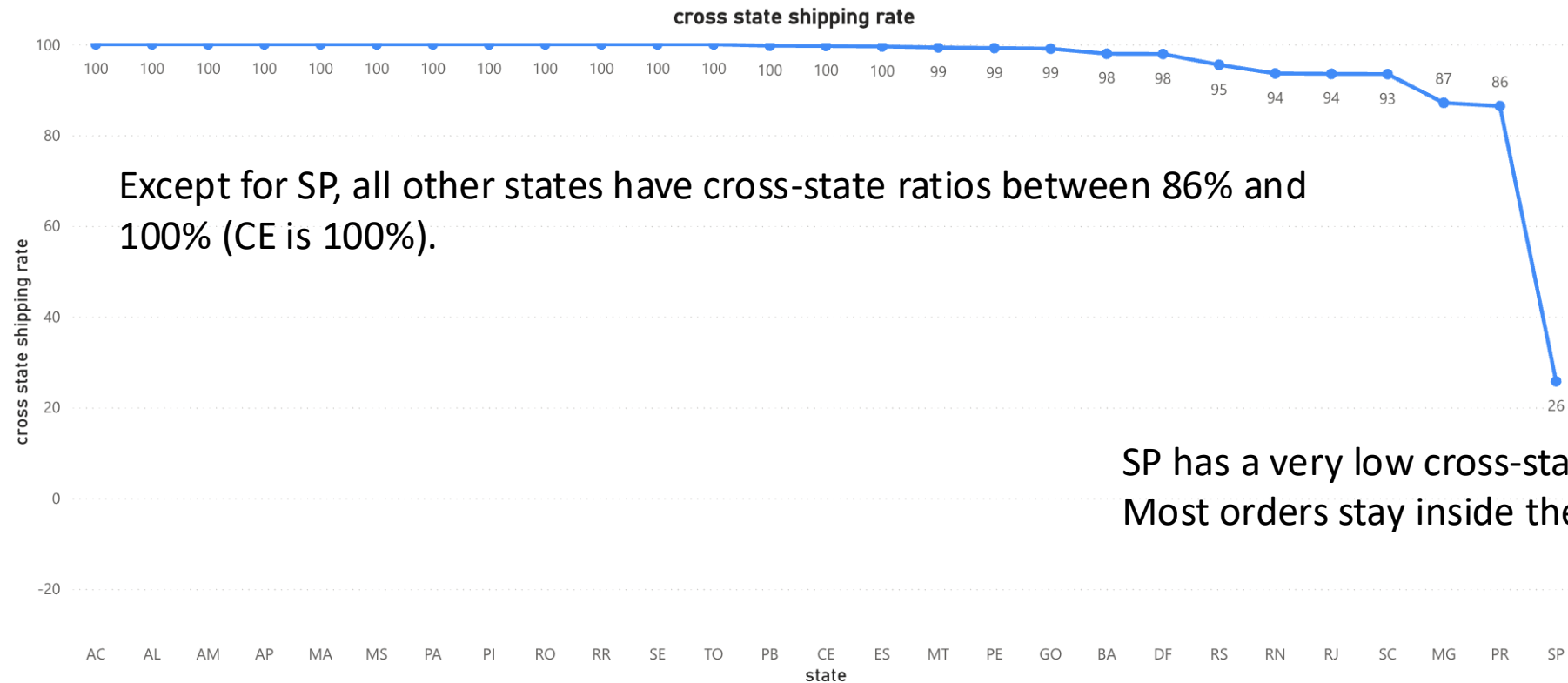
Very Low Repeat Rate: Out of 56,926 mature customers, 96.05% bought only once. Very few people buy again.

Big Gap Between States:

Repeat rates differ by location: SP reaches 4.26%, while distant CE drops to 2.09%—almost cut in half.



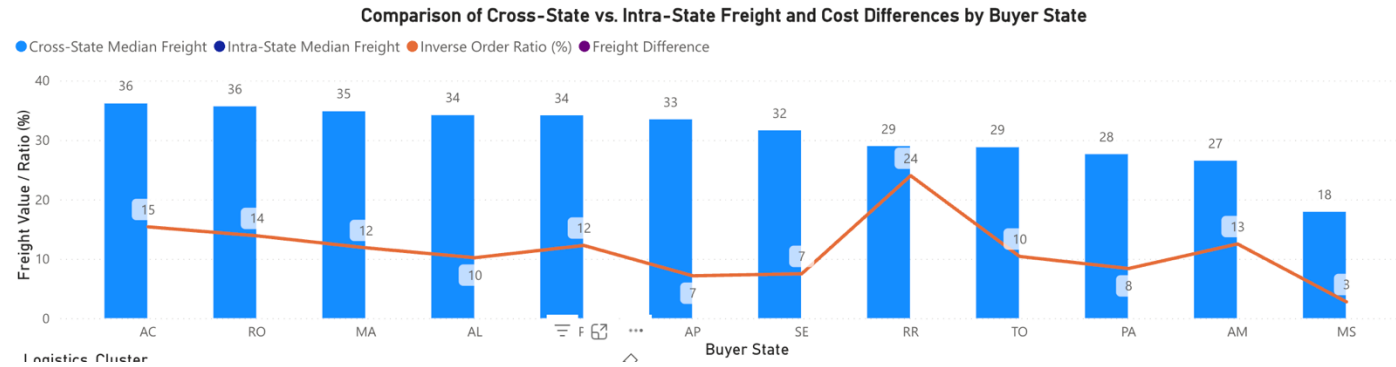
Cross-State Shipping Ratio by State



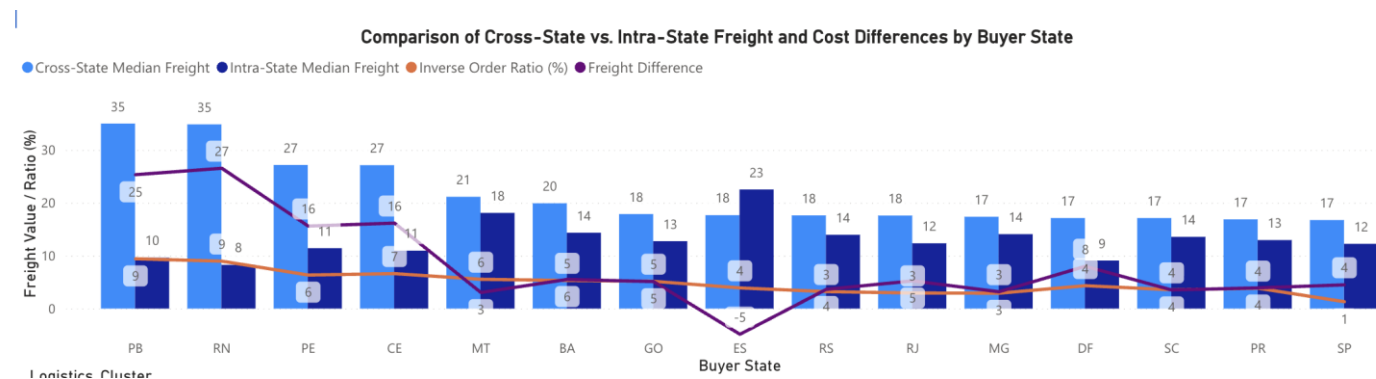
To analyze the data more clearly, states are grouped by their shipping patterns to observe different customer experiences:

- **Pure Cross-State:** Observing remote areas that rely entirely on out-of-state shipping.
- **Mixed Logistics:** Observing states with both local and out-of-state shipping options.

Preliminary Freight Analysis: High Cost Barriers for Out-of-State Buyers

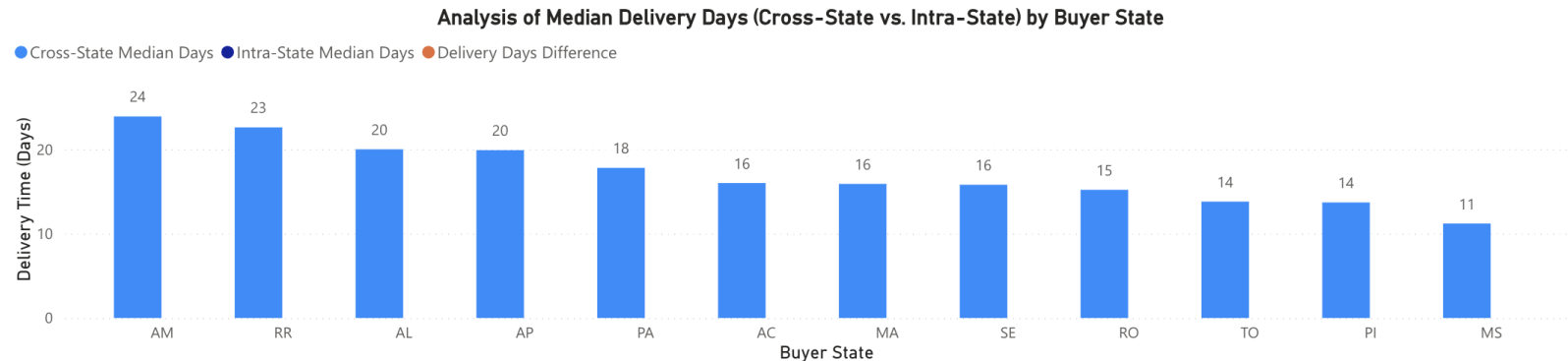


High Fees in Remote States: Remote states pay high fees of 34-36 with zero cheap local options.

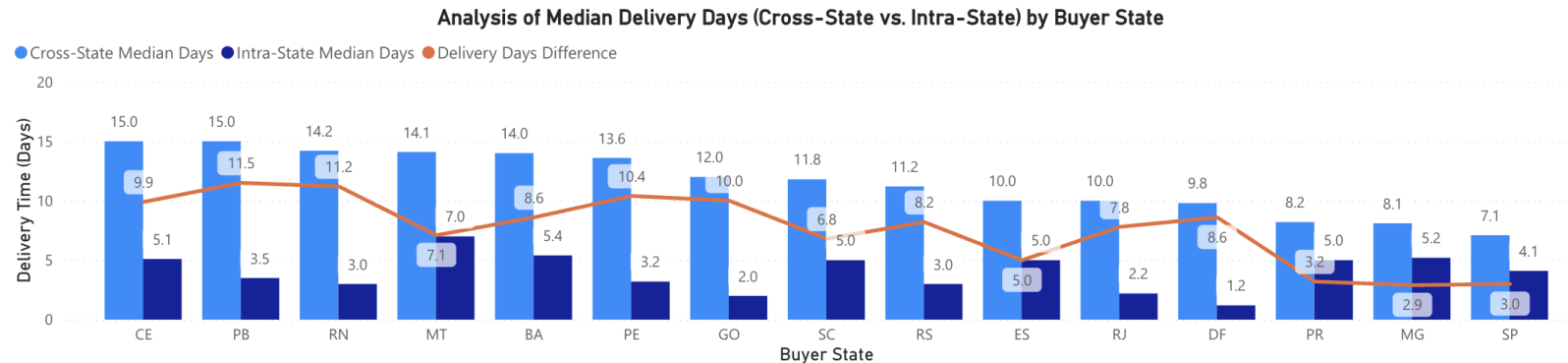


High "Cross-State Tax" in Mixed States: In PB and RN, cross-state shipping is 3-4x more expensive than local (difference of 25-27), while SP has almost no difference.

Preliminary Freight Analysis: High Cost Barriers for Out-of-State Buyers



Extremely Long Wait in Remote States: Remote states face very long wait times of 23-24 days with zero fast options.



Significant Delivery Gap in Mixed States: In PB and RN, cross-state delivery is 3-4x slower than local (difference of 11+ days), while SP has almost no difference.

Preliminary Review Analysis: Hidden Dissatisfaction for Out-of-State Buyers



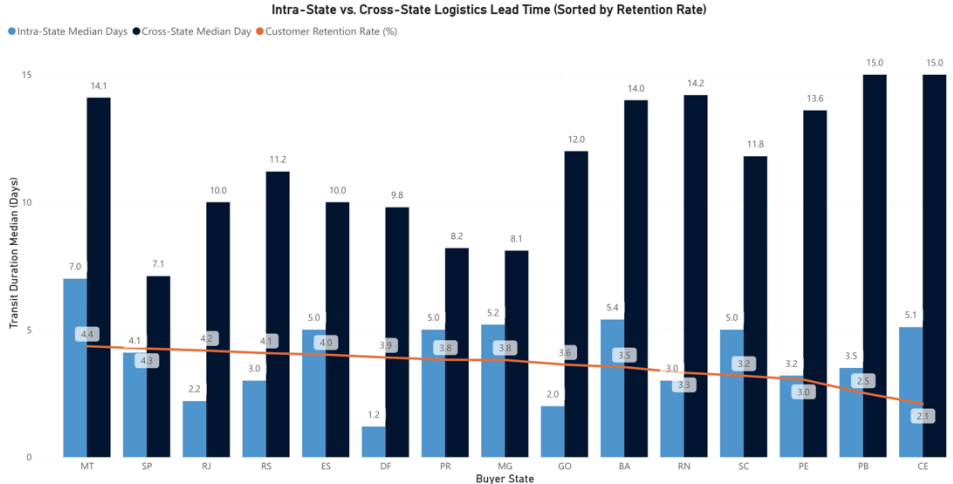
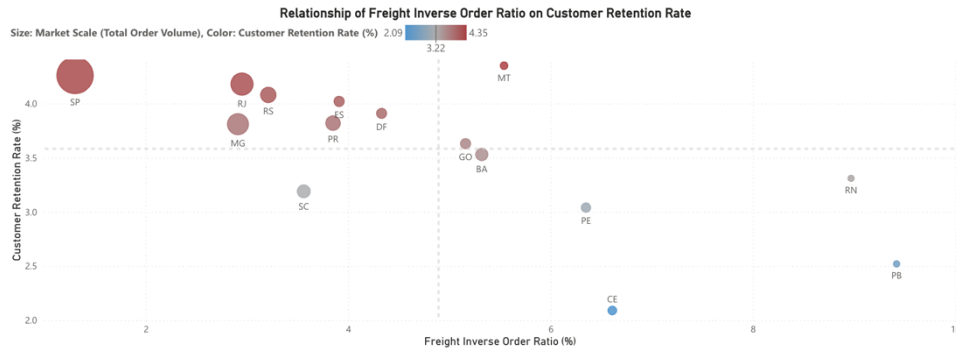
Deceptive Average Scores: Average scores are very close (3.9 to 4.2) across all states, which easily hides real customer pain points.



To avoid misleading averages, future analysis will use "Low-Rating Ratio (1 & 2 stars)" as the main metric to measure satisfaction.

High "Low-Rating Ratio" in Cross-State Orders: Cross-state orders (dark blue line) have much higher low-ratings. Distant CE has a 19% low-rating ratio, while SP is only 11%.

Result Analysis: Mixed Shipping States



The Freight-Reverse Penalty

- **Overall Negative Correlation:** Higher freight-to-product cost ratio overall correlates negatively with customer retention.
- **Cost Inversion & Retention Collapse:** ~10% of orders in PB & RN face cost inversion (freight > product), dragging retention down to its lowest levels .

Logistical Lead Time & Transit Gaps

- **The 10-Day Delivery Gap:** Low-performing PE/CE /PB face 15-day cross-state waits—over 10 days slower than local shipping (3-5 days).
- **Slower Delivery, Lower Retention:** Regional retention drops steadily as the transit gap between local and cross-state shipping widens.

MT Exception

- **High Friction, High Loyalty:** Despite high freight costs and 14.1-day delays, MT leads retention at 4.4%, **proving a strong captive market with rigid demand.**

Improvement Plan

Group 1: RN, PE, PB, CE (High Freight Costs, Slow Delivery)

- **Give Free Shipping:** Offer free shipping to stop customers from paying more for delivery than the product itself.
- **Speed Up Delivery:** Work with faster shippers to cut down the long 15-day waiting time.

Group 2: RJ & RS (Good Repeat Rates, But Can Be Better)

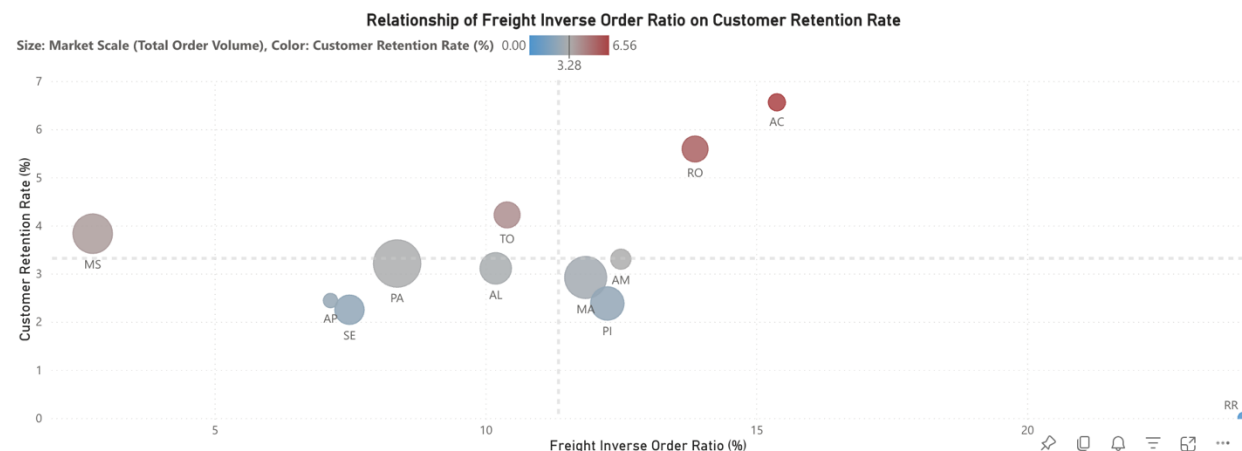
- **Keep Them Happy:** These states already have good repeat buyers, but we must not let them wait too long.
- **Deliver Faster:** Improve cross-state shipping speeds so customers don't have to wait so many days.

Group 3: MT (Mato Grosso — Loyal Buyers with Real Needs)

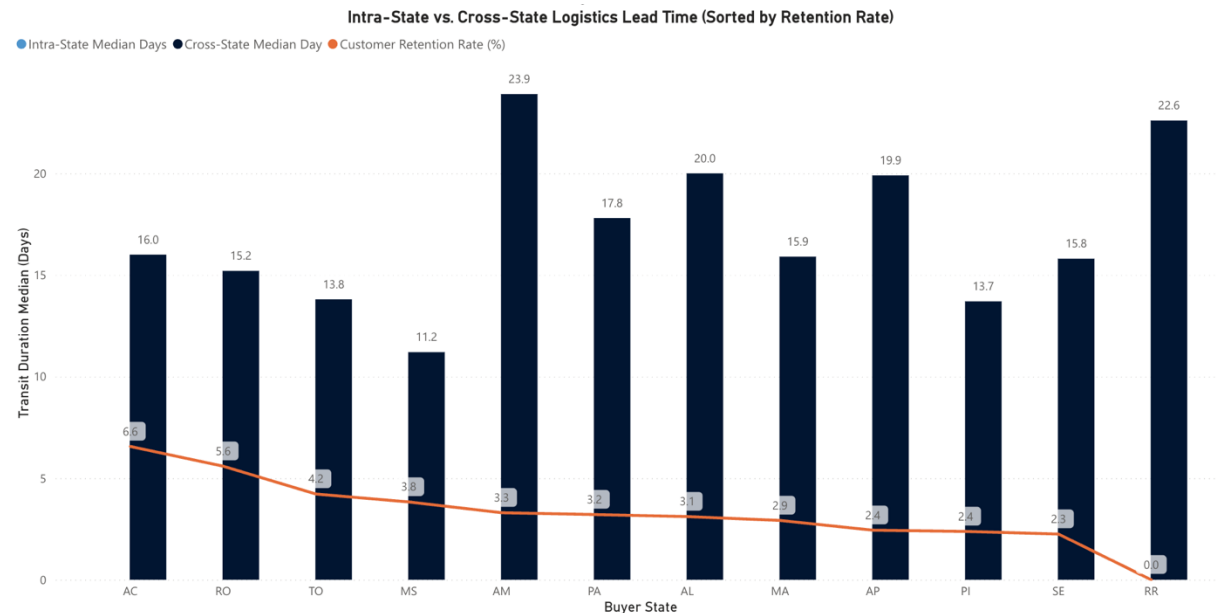
- **Keep Popular Items in Stock:** Always have enough stock for the specific products they need and buy most, such as **health & beauty** and **sports & leisure** products.
- **Offer Shipping Discounts:** Give them special shipping discounts to thank and keep these loyal, high-value customers.

state	product_category	total_purchased
MT	health_beauty	90
MT	sports_leisure	76
MT	furniture_decor	75
MT	bed_bath_table	70
MT	telephony	66
MT	watches_gifts	61
MT	auto	56
MT	computers_accessories	55
MT	garden_tools	53
MT	housewares	52

Result Analysis: Pure Cross-State Shipping States



The AC & RO Growth Opportunity: They face high shipping costs but average waiting times. Surprisingly, their repeat rates are outstanding (6.56% and 5.59%).



Delivery Speed is Key: Delivery speed matters more than the freight cost ratio. **PA and AL** have similar or better freight cost ratios than TO, but much longer wait times (18–20 days vs. TO's 13.8 days). This long wait drags their repeat rate down to ~3%.

Improvement Plan

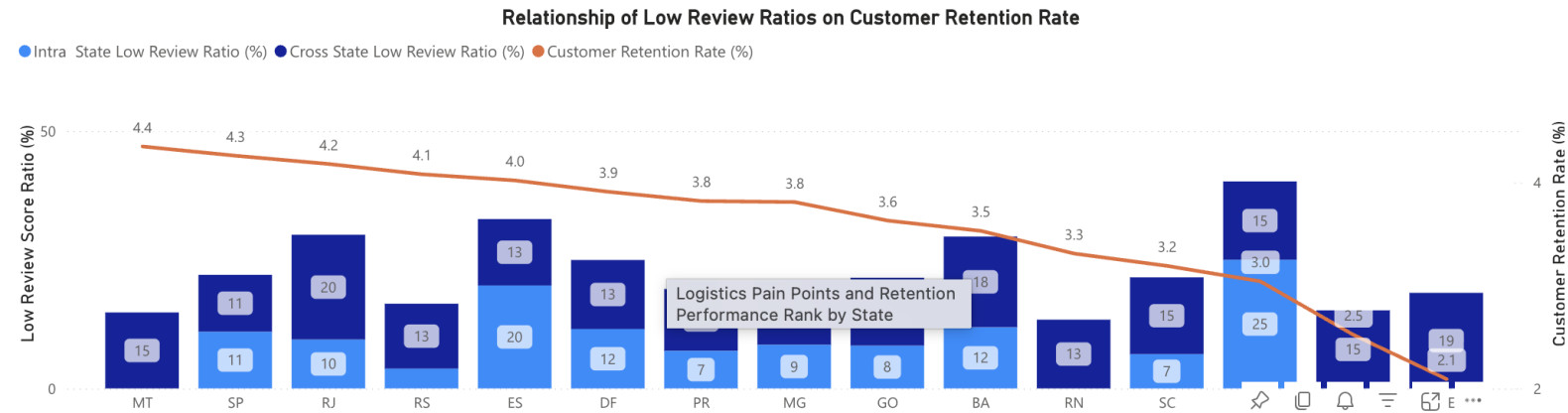
Group 1: AC & RO (High Costs, High Loyalty)

- **Give Shipping Discounts:** Since they are already highly loyal, lowering their shipping costs will easily get them to buy much more.

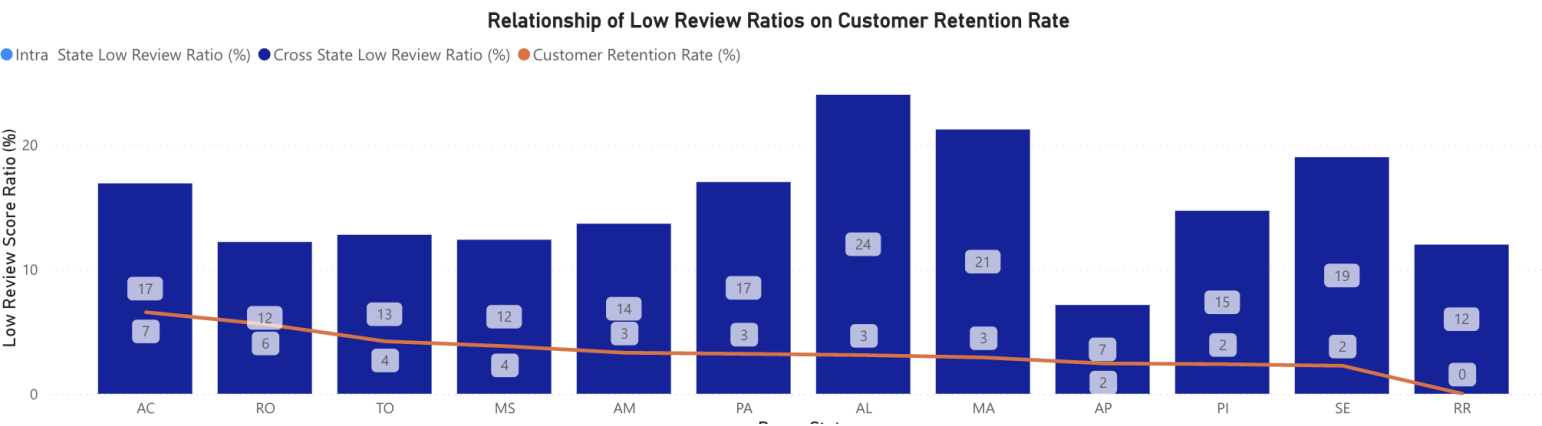
Group 2: PA & AL (Slow Speed, Lower Repeat Rates)

- **Deliver Faster:** Work with faster shippers to cut down their long 18–20 day waiting times closer to TO's 13-day level.

No Clear Link Between Bad Reviews & Retention



- States with high bad reviews often have better repeat rates.
- A bad review may also be tied to a specific product's quality or a single seller's service, rather than just the platform's shipping experience.



04. Conclusions

Conclusion

- From an e-commerce perspective, resolving shipping-related pain points is the most actionable and effective way to maintain or boost customer repurchase intent.
- By assessing each state's current performance, we can implement regional solutions and unlock high-potential markets:
 1. AC & RO (High-Loyalty Potential): They maintain outstanding repeat rates (~6%) despite high costs. Lowering their shipping barriers with discounts will easily capture more high-value demand.
 2. PA & AL (Slow Speed Bottleneck): Long 18–20 day waits drag repeat rates down to ~3%. Improving logistics speed is critical to stop customer loss.

Power BI link:

<https://app.powerbi.com/groups/me/reports/ff24b445-d81a-42b7-91c9-d3b3de96930a/f99be58fba1f45a59d78?experience=power-bi>